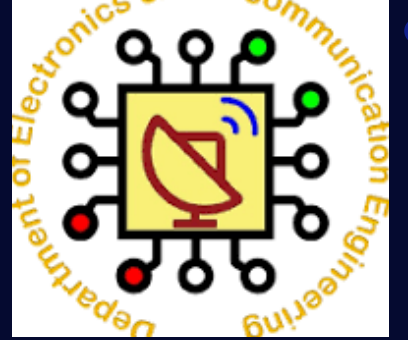




Bharatiya Vidya Bhavan's
Sardar Patel Institute of Technology
(An Empowered Autonomous Institute Affiliated to Mumbai University)
Munshi Nagar, Andheri (W), Mumbai-400058



FPGA-Based Hardware Authentication Framework Using Ring Oscillator PUF

Mentor : Dr. Payal Shah

Course Name: Mini Project I

Course Code : PR1

Semester : V

Academic Year : 2026-27

Team members:

Parinita Chavan (2024200017)

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Khushi Bramhan (2024200011)

MOTIVATION

FPGAs Are Everywhere, and Trust Is Assumed

Modern electronic systems rely on programmable hardware such as FPGAs in aerospace, defense, automotive, industrial automation and communication systems.

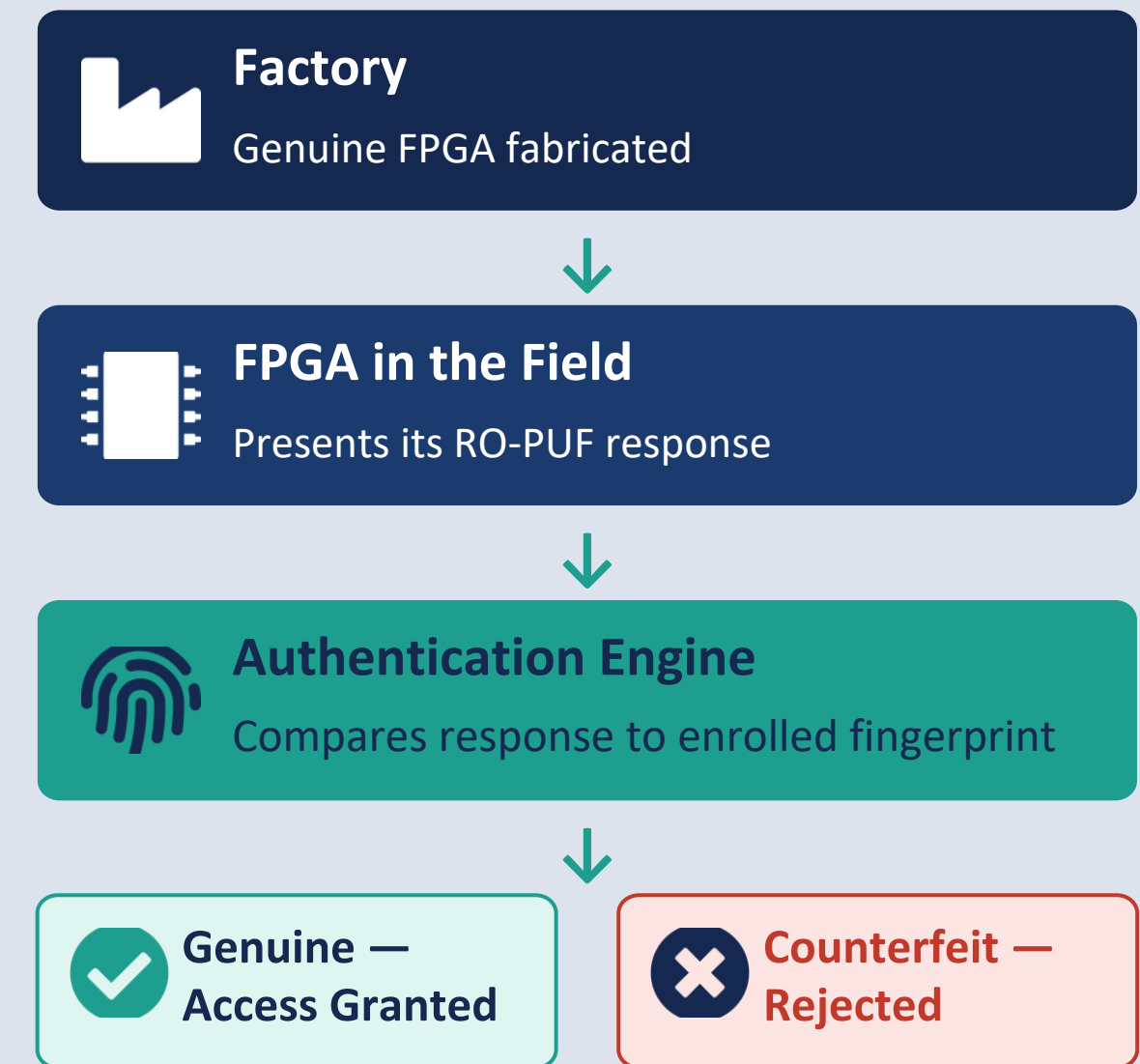
Replaceable Hardware Means Cloneable Hardware

Since FPGAs can be replaced, cloned or counterfeited, there is a growing need to verify the identity of a device before it is trusted.

Silicon Has Its Own Fingerprint

Unlike passwords or stored keys, a Physically Unclonable Function (PUF) generates a unique hardware fingerprint from microscopic manufacturing variations, making each FPGA inherently distinguishable.

DEVICE AUTHENTICATION FLOW



PROBLEM STATEMENT

Existing authentication methods often depend on stored secret keys, which can be copied or extracted from device memory. There is a need for a hardware-based authentication mechanism that uniquely identifies each FPGA device, using a fingerprint that is difficult to clone, requires no permanently stored secret key, and remains reliable despite minor environmental variations.

Stored Keys Can Be Copied

Conventional authentication relies on secret keys stored in non-volatile memory, which can be extracted or duplicated.

Needs to Be Unclonable

The mechanism must uniquely identify a device and resist duplication, even by an adversary with physical access.

Reliable Despite Variation

Authentication must stay consistent across temperature and voltage fluctuations without a permanently stored key.

Design and implement an FPGA-based hardware authentication framework using Ring Oscillator PUF to uniquely identify FPGA devices and evaluate its reliability using statistical performance metrics.

OBJECTIVES

O1

Design the RO-PUF

Design a Ring Oscillator PUF on FPGA using matched oscillator pairs and frequency counters.

O2

Generate a Device Fingerprint

Generate a unique fingerprint for each FPGA from its RO frequency comparisons.

O3

Interface with STM32

Interface the FPGA with STM32 for fingerprint acquisition over UART.

O4

Develop Enrollment Software

Develop Python software for device enrollment and run-time authentication.

O5

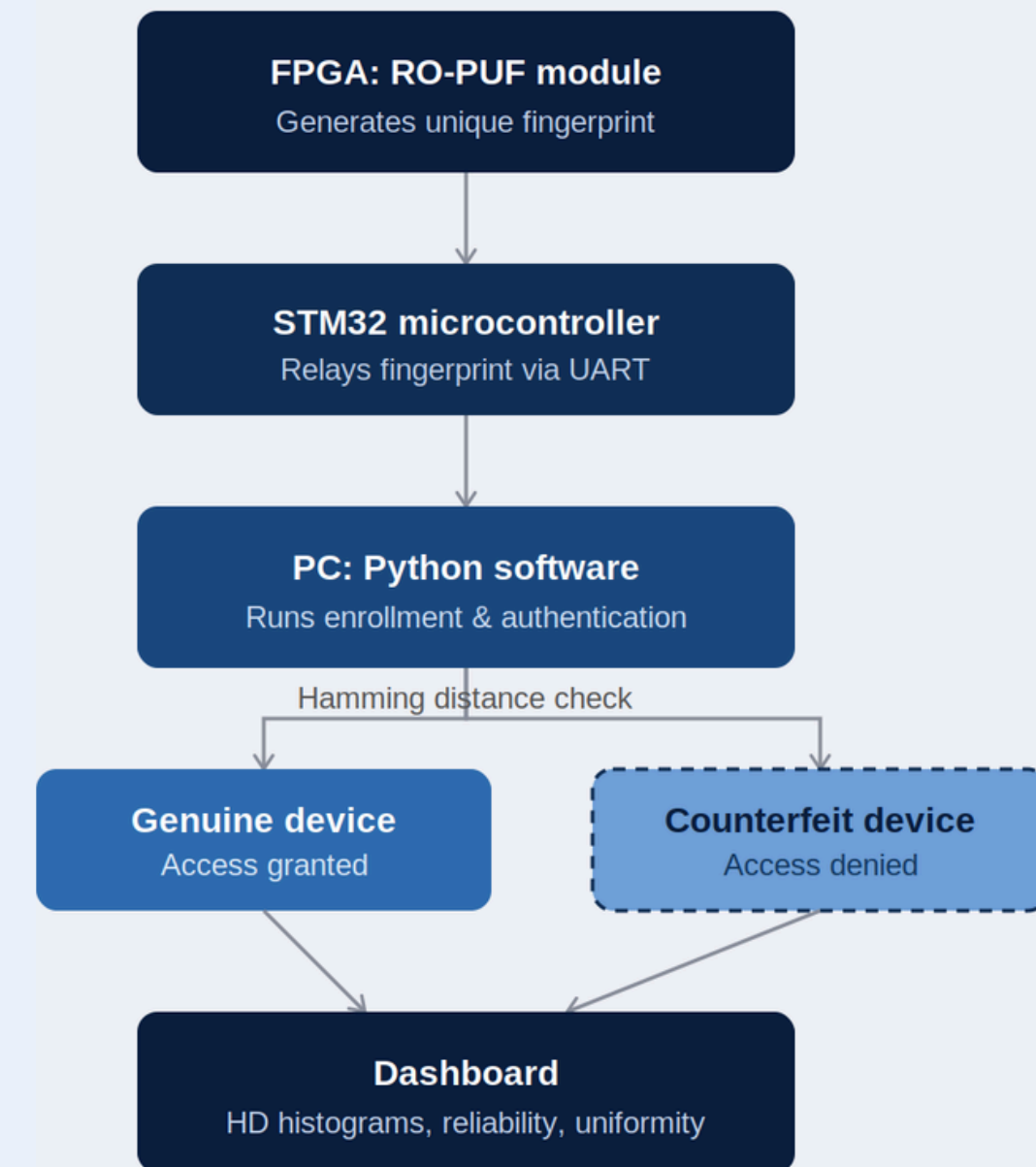
Evaluate Performance

Evaluate authentication performance using Hamming Distance, Reliability and Uniformity.

O6

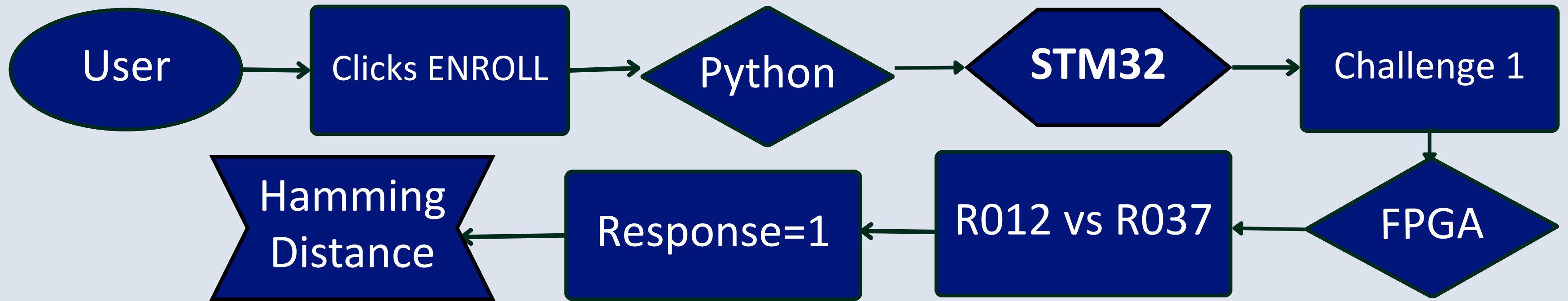
Visualize Results

Visualize authentication results through a graphical dashboard.

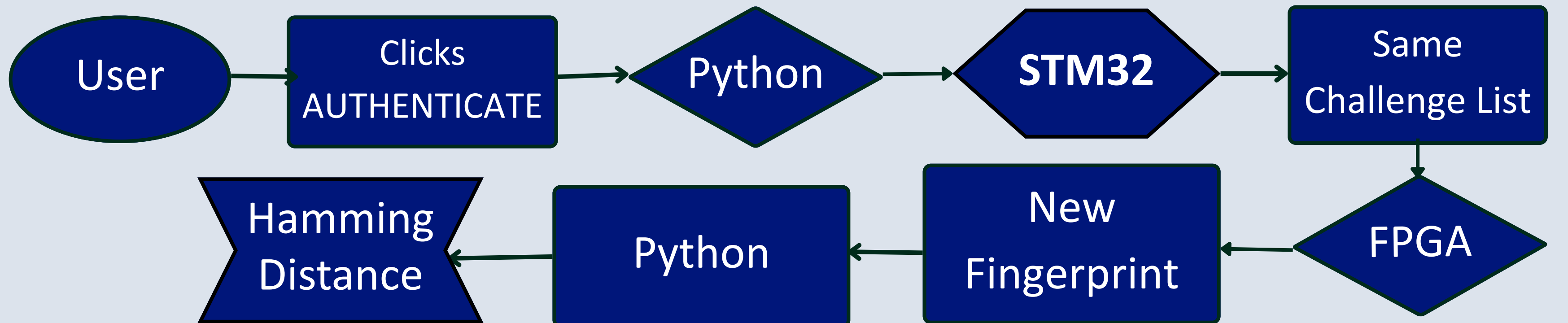


WORK FLOW

ENROLLMENT



AUTHENTICATION



MAPPING WITH THE SUBJECTS LEARNED

Course	How It Maps onto This Project
Digital Electronics	Ring Oscillators, Counters, FSMs
HDL Programming	Verilog implementation of the PUF
Microcontrollers	STM32 interfacing and UART communication
Computer Organization & Architecture	FPGA architecture and hardware modules
Probability & Statistics	Hamming Distance, Reliability, Uniformity
Signal, Network & System	Frequency analysis of Ring Oscillator outputs

REFERENCES

1

- A Novel Configurable Ring Oscillator Based PUF for Hardware-Assisted Security and Recycled IC Detection (IEEE Access, 2024)
- <https://ieeexplore.ieee.org/document/10818478>

2

- Fault-Tolerant Reversible-Logic Based RO-PUF for Secure Device Authentication (IEEE TCAS-I, 2024)
- <https://ieeexplore.ieee.org/document/10601312>

3

- A Novel FPGA Mutually Coupled Configurable Ring Oscillator PUF (IEEE AsianHOST 2024)
- <https://ieeexplore.ieee.org/document/10838475>

4

- Configurable RO-PUF with Improved Thermal Stability for Lightweight Applications (IEEE ICM 2024)
- <https://ieeexplore.ieee.org/document/10815902>

Thank You